



William Kirby- an English Biologist, famous for his enormous work in Entomology. He is called the '**Founder of Entomology**'.

Unit



Periplaneta americana (Cockroach)

Cockroach belongs to the phylum Arthropoda and the class Insecta. Insecta is the largest class of animals in the Kingdom –Animalia. These are terrestrial animals but adapted to live in all habitats, except in deep waters of the seas. Insects are distinguished from other arthropods in possessing three **tagmata** in the body and three pairs of legs. The study of insects is called **Entomology**.

Cockroach is an abundantly and easily available insect. It also exhibits all the characteristic features of the class **Insecta**. Besides this, its large size of the body makes it a convenient specimen, for dissection and demonstration. Therefore it is an excellent example for the study of insects.

Cockroach is a common household pest that contaminates our food with its excreta. It can transmit a number of bacterial diseases by contaminating food materials. These are mostly the inhabitants of the tropical countries. However some species are found in the temperate regions. The two common species of cockroaches found in India are: *Blatta orientalis* and ***Periplaneta americana***.

Periplaneta americana - type study

7.1 HABITAT AND HABIT

- 7.1 Habitat and habit
- 7.2 External features (Morphology)
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Periplaneta americana, was a native of the tropical America. Now it has become a *cosmopolitan* insect. The name *Periplaneta americana* was coined by **Burmeister**. It is usually found in kitchens, hotels, bakeries, warehouses, etc., It is a **nocturnal** insect (it comes out of its hiding places to feed at night times). During the day time it remains inactive and takes rest. Its flattened form of the body enables it to enter small crevices in the floor or walls. It is an **omnivorous** insect, eating all types of food. It locates its food by the sense of smell. *Periplaneta* is a **cursorial** (swift runner) animal.

Classification

Phylum - Arthropoda
Class - Insecta
Order - Orthoptera
Family - Blattidae
Genus - *Periplaneta*
Species - *Americana*

Why do cockroaches take rest during day time?

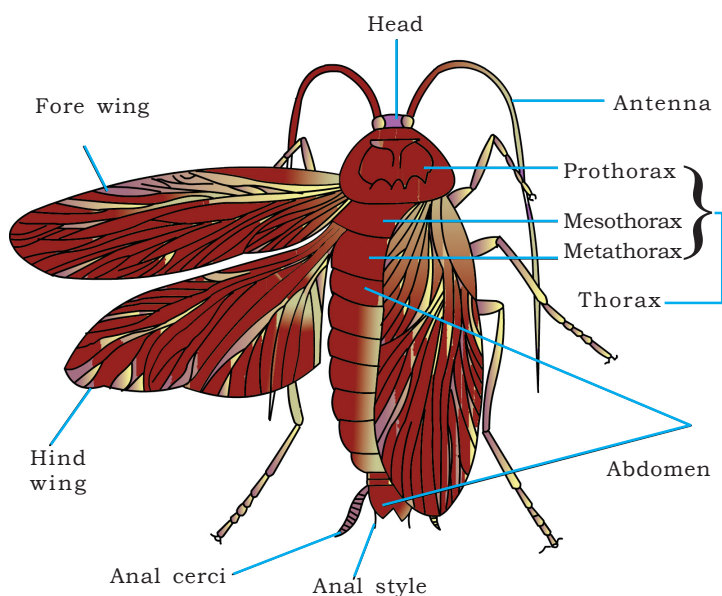


Fig. 7.1 *Periplaneta* (Male): External features

7.2 EXTERNAL FEATURES (MORPHOLOGY)

Cockroach has an elongated, narrow, elliptical, dorso-ventrally depressed body. It exhibits bilateral symmetry. The wings extend beyond the tip of the abdomen in the males. The colour is reddish brown with a light yellow area around the edges of the pronotum. The complete body is externally covered by a **chitinous cuticle** which constitutes the **exoskeleton**. The cuticle is secreted by the underlying cells of the

hypodermis^[1]. It is coated externally by a thin layer of wax, impermeable to water. The **oenocytes**, are believed to secrete **wax**. The cuticle protects the body, prevents loss of water, provides rigidity and offers place for the attachment of muscles. It consists of small plates or **sclerites**, which are joined by soft, flexible **arthrodial membranes**.

1. Body Divisions: The body of cockroach is distinctly segmented and consists of three **tagmata** (morphologically distinct parts of the body) namely head, thorax and abdomen.

A. Head: The head of cockroach is small and triangular. It is called **hypognathous** because it lies hanging almost at right angles to the body with the posterior wider part upwards, and the mouthparts directed down-wards.

The head of cockroach is formed by the fusion of six embryonic segments. It is movably attached to the thorax by a short neck or **cervicum**. It is covered by a number of sclerites which fuse to form a capsule. The top of the head between the eyes is called **vertex**. The vertex has two sclerites called '**epicranial plates**' connected by an '**epicranial suture**'. Below the vertex, the sclerites covering the head in front are – a large **frons**, a narrow rectangular **clypeus** and a movable **labrum**. Covering the sides of the head, below the compound eyes are the 'cheek sclerites' or '**genae**'. At the back of the head capsule there is a large opening called **occipital foramen**. It is bordered by a sclerite called **occiput**. The occipital foramen forms a passage for the oesophagus, aorta, nerve cord and tracheae. At the base of each **antenna**, a small whitish speck called **fenestra** or 'ocellar spot' or 'simple eye' is present. Appendages are absent in the first and third segments of the head. The second segment bears a pair of long, slender and segmented **antennae**, one on each side of the head. The antennae are tactile and olfactory

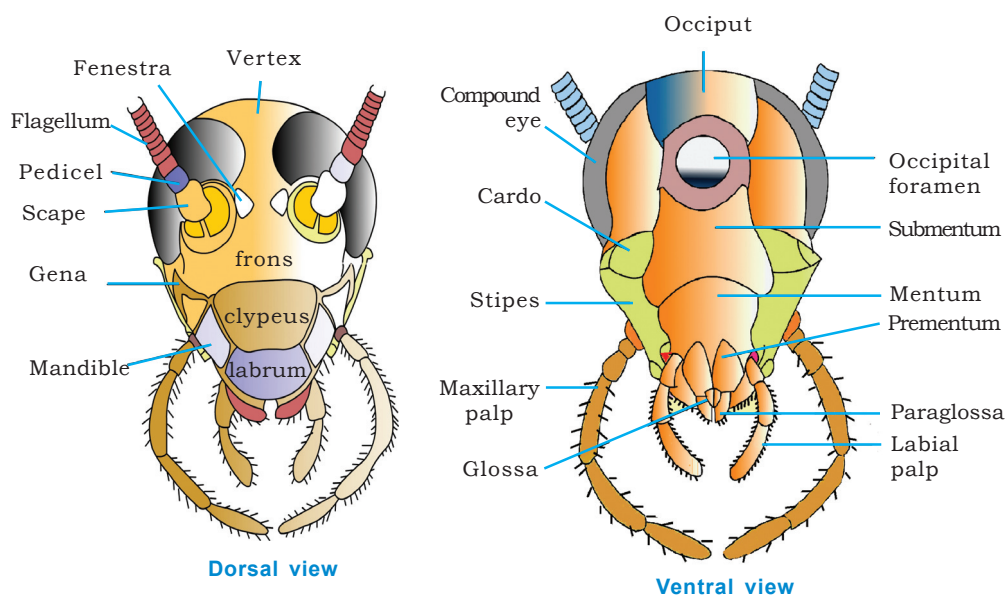


Fig. 7.2 *Periplaneta*: Head dorsal and ventral View

^[1] It is the epidermis below the cuticle, hence called hypodermis

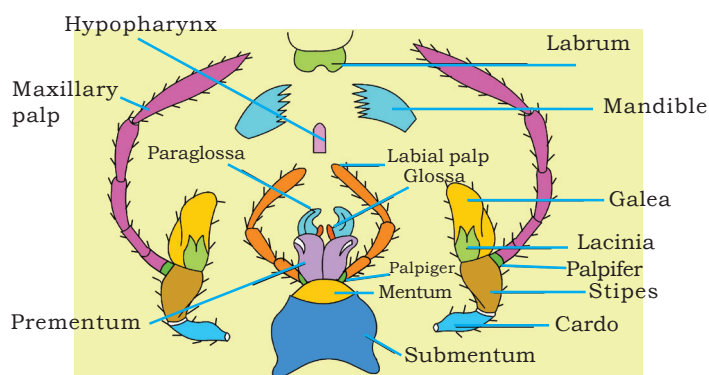


Fig. 7.3: *Periplaneta*: Mouth parts

in function. The fourth segment bears a pair of **mandibles**. The fifth segment has a pair of '**first maxillae**'. The sixth segment bears a pair of '**second maxillae**', which fuse to form the **labium** (also called 'lower lip').

Mouthparts

The mouthparts of *Periplaneta* are of **Biting** and **Chewing** type. These are the movably articulated appendages surrounding the mouth.

They include a pair of mandibles, a pair of first maxillae, a pair of second maxillae or labium and a tongue or **hypopharynx** and an upperlip, the labrum. They enclose a space called the '**preoral cavity**'.

Labrum (upper lip)

The labrum is a vertical, rectangular plate, movably connected with the clypeus. It forms the anterior wall of the preoral cavity and is also called the 'upper lip'. It bears '**gustatory sensillae**' on its inner surface. The labrum serves to hold the food and helps in tasting it.

Mandibles

A pair of triangular, hard, unjointed, chitinised mandibles is present on the sides of the mouth, connected with the genae. The inner margins of mandibles have teeth like structures. These teeth help to masticate the food. Two pairs of muscles called **adductor** and **abductor** muscles help in the movement of the mandibles. Each mandible has incising and grinding teeth.

First maxillae

A pair of first maxillae is situated on the sides of the mouth immediately behind the mandibles. Each maxilla is 'biramous' and consists of three parts: the protopodite, endopodite, and the exopodite. **Cardo** and **stipes** constitute the protopodite. A five jointed maxillary palp, arising from a sclerite called **palpifer**, on the outer side of the stipes, constitutes the exopodite. The maxillary palps are used for cleaning the antennae and the front pair of legs. From the distal end of the stipes, internal to the maxillary palp, arises the endopodite. Each endopodite has two parts, an outer broad hood-like **galea** and inner pincer-like (resembling a forceps) **lacinia**. Maxillae serve to hold the food and bring it to the mandibles.

Second maxillae (Labium or Lower lip)

The second pair of maxillae lies behind the first pair. Both the second maxillae fuse to form a broad plate like **labium** or **lower lip**. It consists of three parts: the upper

submentum, the middle mentum and the lower prementum. A three segmented labial palp, arising from a sclerite called **palpiger**, is present one on each side of the prementum. It prevents the food from falling, and pushes it into the preoral cavity.

The distal end of the prementum bears a pair of '**paraglossae** (comparable to **galeae** of the first maxilla), a pair of '**glossae**' (comparable to **laciniae** of the first maxilla). The glossae and paraglossae together constitute the **ligula**.

Hypopharynx (Tongue or Lingua): The hypopharynx is a rod like, grooved chitinous structure, hanging in the preoral cavity between the first maxillae. The hypopharynx is also called 'tongue' or '**lingua**'. It bears the opening of the '*efferent salivary duct*'.

Note: Biting and chewing type of mouth parts are believed to be the most primitive type (as most larvae of insects possess such type of mouth parts).

Why are the mouthparts of cockroach called biting and chewing type?

Neck (cervicum)

Neck is short, slender, flexible tube (a part or extension of the thorax) that connects the head with the thorax. It is supported by **four cervical sclerites**. It is not a 'tagma' of the body. It helps in the movement of the head.

B. Thorax: The thorax is the second 'tagma' of the body. It consists of three segments: prothorax, mesothorax, and metathorax. Each segment of the thorax is covered by four sclerites -one on the dorsal side (called **tergum** or **notum**), one on the ventral side (called **sternum**) and two (called **pleura**), on the lateral sides. The tergum of the prothorax is called **pronotum**. It is the largest sclerite of the body and covers the neck and a little part of the head. The terga on the mesothorax and metathorax are called **mesonotum** and **metanotum** respectively. The sclerites of each segment are joined by thin, soft, flexible membranes called arthrodial membranes, which permit movement, between the segments.

Legs: There are three pairs of jointed, walking legs in cockroach, attached one pair to each thoracic segment on the ventral side. The legs are connected with the pleura and sterna of the thoracic segments. According to the segment that bears them, they are called prolegs, mesolegs and metalegs. All the legs are similar in structure. Each leg is made up of five segments or **podomeres**. The different podomeres (from the base to the tip) are serially: **coxa**, **trochanter**, **femur**, **tibia** and **tarsus**.

The coxa connects the leg with the thorax. The trochanter is *movably attached* to the coxa, and *fused* with the femur. The femur and tibia bear chitinous bristles. The joints of tarsus are **tarsomeres** which bear soft pads called **plantulae** on their inner surface. The terminal joint of the tarsus, ends in a pair of **claws**. Between the claws there is a soft hairy pad called **pulvillus** or **arolium**.

Wings: Two pairs of wings, the fore wings and the hind wings, are present in cockroach. The forewings are thick, opaque and leathery. They do not help in flight, but cover and protect the hind wings when they are not in use. They are called **tegmina** (singular: **tegmen**). The hind wings are broad, thin, transparent and delicate.

They help in flight and remain folded below the tegmina when not in use. The wings of cockroach contain a network of hollow **veins** or **nervures**.

C. Abdomen: The abdomen consists of **ten** segments. Each segment is covered by the dorsal tergum, the ventral sternum and the two lateral pleura or pleurites. There are ten terga but only nine sterna as the tenth sternum is absent. The eighth tergum in the male and both eighth and ninth terga in the female are not visible as they are overlapped by the seventh tergum. The tenth tergum extends beyond the posterior end of the body and has a deep notch / groove in the middle of its free end. In the male nine sterna are visible whereas in the female, only seven sterna are visible. The seventh, eighth and ninth sterna together form a **brood pouch**. The brood pouch has two parts the anterior **genital chamber** or **gynatrium** and posterior **oothecal chamber**.

The posterior end of the abdomen has a pair of **anal cerci**, a pair of **anal styles** and gonapophyses in the males. Anal cerci are jointed (15 segmented) and arise from the lateral sides of the tenth tergum and are found in both the sexes. The anal styles are without joints and arise from the ninth sternum (seen only in the males). The **gonapophyses** are small chitinous processes arising from the ninth sternum in the males and eighth and ninth sterna in the females. They are the external genital organs. The **anus** is at the posterior end of the abdomen in the 10th segment. The genital aperture in male is present just below the anus on one of the **gonapophyses** and in female it is located on the **eighth sternum**.

Body wall: The body wall consists of three layers: **cuticle**, **epidermis** or **hypodermis** and **basement membrane**. The cuticle is the outer most layer. It is again differentiated into three layers, namely, an outer thin, waxy **epicuticle**, middle thick **exocuticle** made of tough pigmented chitin and an inner much thick **endocuticle**, made of soft layers of **chitin**. The articular membranes lack the exocuticle and so they are flexible. The epidermis lies below the cuticle. It comprises a single layer of columnar cells resting on the basement membrane. It secretes the cuticle. The basement membrane is the inner most layer of the body wall. The cuticle protects the delicate internal organs and prevents the loss of water by evaporation and provides surface for the attachment of muscles.

How does the cuticle help in preventing the loss of water?
What makes the articular membrane flexible?

Body cavity and coelom

The body cavity / perivisceral cavity of *Periplaneta* is not a true coelom and it is not lined by peritoneum. It is called '**haemocoel**' as it is filled with blood or haemolymph. The schizocoelom is restricted to the spaces around the reproductive organs.

Why is the body cavity of a cockroach not a true coelom but a haemocoel?

Fat bodies

The haemocoel contains many large sized fat bodies. These bodies are also called **corpora adiposa**. These are similar to the liver of the vertebrates in certain functions. A fat body has many lobules containing four types of cells namely **trophocytes** (store food) **mycetocytes** (contain symbiotic bacteria), **oenocytes** (synthesise and store lipids), **urate cells** (store uric acid).

7.3 LOCOMOTION

Periplaneta has double mode of locomotion. 1-Running (cursorial locomotion) 2-Flying.

1. Running or cursorial locomotion

Cockroach is a cursorial insect and it can run on the ground with the help of legs. During walking or running cockroach forms two **tripods** with the legs. Each tripod is formed by foreleg and hind leg of one side and middle leg of the other side. During locomotion, the three legs of one tripod are kept on the ground and the other three legs of another tripod are carried forward. In this process, the foreleg and the hind leg of the tripod pull and push the body while the middle leg acts as a **pivot**. Thus cockroach moves using the two tripods alternately. The animal does not slip back as plantulae, pulvilli and claws provide grip. The claws and the arolium help in locomotion on **rough surfaces** whereas plantulae are useful on **smooth surfaces**.

How is each tripod of cockroach formed during locomotion?

2. Flying: Though cockroach is a cursorial insect, it can fly over short distances with the help of its wings. While flying the first pair of wings is stretched out at right angles to the body. The second pair of wings is moved up and down during flight. Wings are elevated by the contraction of dorsoventral muscles. Contraction of the dorso-longitudinal muscles and relaxation of the dorsoventral muscles depresses the wings.

7.4 DIGESTIVE SYSTEM OF PERIPLANETA

The digestive system of cockroach consists of an alimentary canal and the associated glands. The preoral cavity, surrounded by the mouth parts, is present in front of the mouth. The hypopharynx divides it into two chambers called **cibarium** (anterior) and **salivarium** (posterior).

I. Alimentary canal

The alimentary canal of cockroach is a long tube and is coiled at some places. It extends between the mouth and the anus. It is divided into three regions, namely, foregut

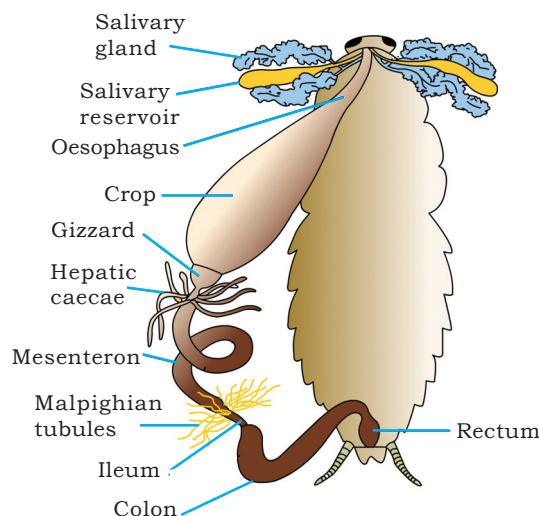


Fig. 7.4 *Periplaneta* : Digestive system

or **stomodaeum**, midgut or **mesenteron** and hindgut or **proctodaeum**. The foregut and hindgut are internally lined by ectoderm. The mesenteron is lined by the endodermal cells.

1. Foregut or stomodaeum

The foregut includes pharynx, oesophagus, crop, and gizzard. It is internally lined by a chitinous cuticle. Mouth opens into the **pharynx**, which in turn leads into a narrow tubular oesophagus. The **oesophagus** opens behind into a thin walled distensible sac called **crop**. The crop serves as a reservoir for storing food. Its outer surface is covered by a network of tracheae.

Behind the crop there is a thick walled muscular **proventriculus**, or **gizzard**. The chitinous inner lining of the gizzard has six powerful teeth, which form an efficient grinding apparatus. Behind each tooth is a hairy pad, which bears backwardly directed bristles. Among these plates, food is thoroughly ground into fine particles. These food particles are filtered by the bristles. The gizzard thus acts both as a **grinding mill** and also as a **sieve**. There is a membranous projection of the gizzard into the mesenteron in the form of a funnel called **stomodeal valve**. This valve prevents the entry (regurgitation) of food from the mesenteron back into the gizzard.

2. Midgut (mesenteron or ventriculus)

The midgut is a short and narrow tube behind the gizzard. It is also called **mesenteron** or **ventriculus**. Between the ventriculus and the gizzard, arising from ventriculus, there are six to eight finger like diverticula called **hepatic caecae**. They are helpful in digestion and absorption of the digested food materials. Ventriculus is functionally divided into an anterior

The secretory part of the ventriculus has many gland cells and it secretes several enzymes. The 'bolus' of food in the mesenteron is enveloped by a chitinous and porous membrane called **peritrophic membrane**, which is secreted by the mid gut. The peritrophic membrane is a network of chitin fibrils in a glycoprotein matrix, secreted by cells in the anterior region of the midgut.

Digested food is absorbed into the blood through the peritrophic membrane in the posterior absorptive region of the ventriculus. The peritrophic membrane protects the wall of the ventriculus from hard food particles in the food. The opening of the ventriculus into the hindgut is controlled by a **sphincter muscle**. It prevents entry (regurgitation) of undigested food and uric acid from the hindgut into the midgut.

Which part of the alimentary canal produces the peritrophic membrane around the food bolus in *Periplaneta*?

3. Hindgut or proctodaeum

The hindgut is a long coiled tube, consisting of three regions namely **ileum**, **colon** and **rectum**. It is internally lined by chitinous cuticle. The ileum that lies behind the mesenteron is a short tube. Six bundles of fine yellow, blind tubules called **Malpighian tubules** open into the ileum near the junction of mesenteron and ileum. Malpighian tubules are excretory in function. Ileum collects uric acid from the malpighian tubules and undigested food from the mesenteron. Ileum opens behind into a long coiled tube called colon. Colon leads into a short and wide rectum, which opens out through the anus. Rectum bears on its inner side six longitudinal folds called rectal papillae. They are concerned with the reabsorption of water from the undigested food.

II. Digestive glands

The digestive glands associated with the alimentary canal of cockroach are salivary glands, hepatic caecae and glandular cells of the mesenteron.

1. Salivary glands

There is a pair of salivary glands attached to the ventrolateral sides of the crop, one on each side. Each salivary gland has two lobes. Each lobe of salivary gland has many lobules called **acini**. Each acinus is a group of secretory cells called **zymogen cells** with a small ductule. The ductules of both the lobes of a salivary gland unite to form a **common salivary duct** on each side.

The two common salivary ducts are joined to form the **median salivary duct**. Between the two lobes of a salivary gland of each side is a sac called **salivary receptacle** that stores saliva. It leads into a **receptacular duct**, or 'reservoir duct'. The receptacular ducts of both the sides are united to form a **common receptacular duct**, or 'common reservoir duct'. The median salivary duct opens into the common receptacular duct. Later these two form an **efferent salivary duct**. The efferent salivary duct opens at the base of the hypopharynx. Acinar cells secrete saliva, which contains starch digesting enzymes such as **amylase**.

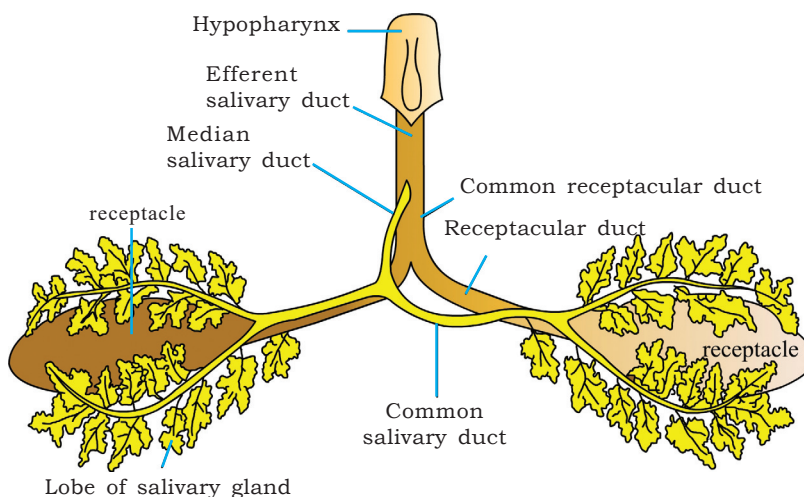


Fig. 7.5 *Periplaneta* : Salivary glands

2. Hepatic caecae

The hepatic caecae are also termed 'midgut caecae'. They contain secretory and absorptive cells.

3. Glandular cells of the mesenteron

The glandular cells of the mesenteron secrete enzymes such as **maltase**, **invertase**, **proteases** and **lipase**.

III. Physiology of Digestion

a) **Food collection:** Cockroach is an **omnivorous** insect. It feeds on all types of organic matter. It locates the food by the olfactory sensillae of antenna, labial palps and maxillary palps. The food is seized with the help of forelegs, labrum and labium. It is passed to the mandibles for biting and chewing. The laciniae, galeae, glossae and paraglossae hold the food during chewing. The labrum and labium prevent the food from falling down. The food is mixed with saliva during mastication.

b) **Digestion:** After swallowing, the food passes through the pharynx and oesophagus, and reaches the crop. In the crop, food is mixed with digestive juices that are regurgitated into it through the grooves of the gizzard. Hence, most of the food is digested in the crop. The partly digested food is filtered by the bristles of the gizzard and later it passes through the stomodeal valve into the ventriculus.

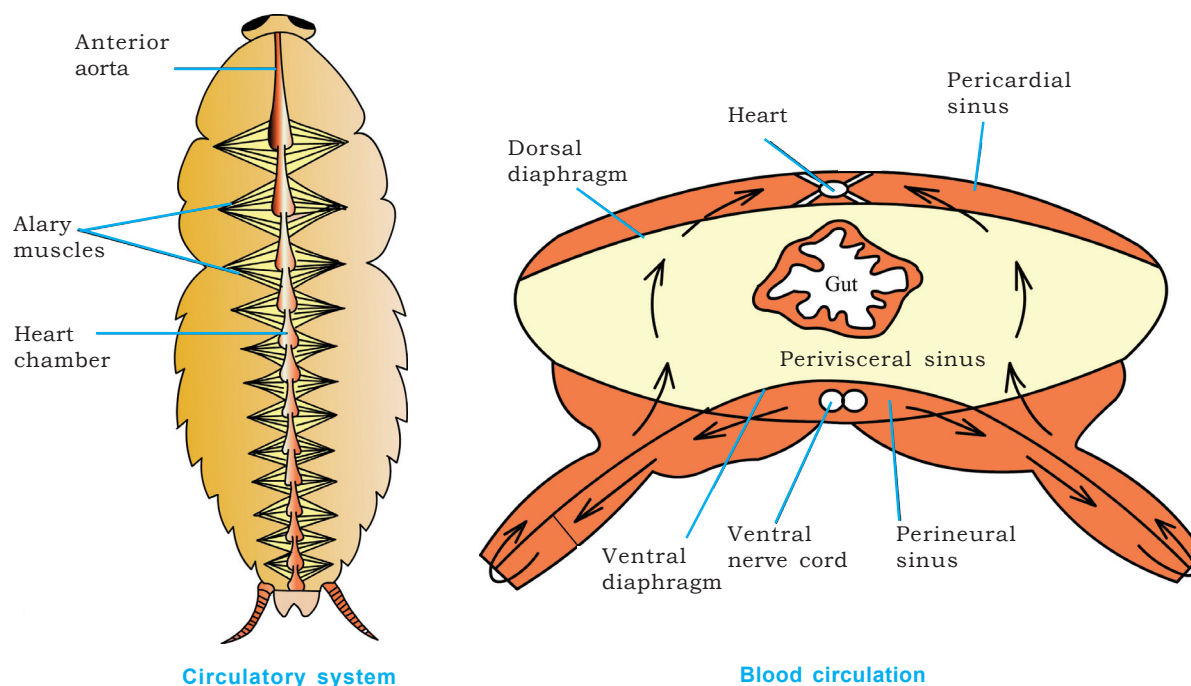
The enzyme amylase of the salivary juice converts starches into **disaccharides**. **Invertase** or sucrase digests sucrose into **glucose** and **fructose**. Maltase converts maltose into **glucose**. The enzyme lipase digests lipids into **fatty acids** and **glycerol**. Proteases digest proteins into amino acids. Cellulose of the food is digested by the enzyme **cellulase** secreted by the microorganisms present in the hindgut of cockroach. Cellulose is converted into glucose.

In the ventriculus, the digested food is absorbed. The undigested food is passed into the ileum, colon, and then reaches the rectum, where water is reabsorbed by rectal papillae. Then the remaining material is finally **defaecated** as **dry pellets**, through the anus.

The crop of *Periplaneta* does not secrete any digestive enzymes, but most of the food is digested in it. Explain how it is possible?

7.5 CIRCULATORY SYSTEM OF *PERIPLANETA*

The circulatory system helps in the transportation of digested food, hormones etc., from one part to another in the body. *Periplaneta* has an **open type** of circulatory system as the blood, or haemolymph, flows freely within the body cavity or haemocoel. Blood vessels are poorly developed and open into spaces. Visceral organs located in

Fig. 7.6 *Periplaneta* : Circulatory system

the haemocoel are bathed in the haemolymph. The three main parts associated with the blood circulatory system of *Periplaneta* are the haemocoel, heart, and blood.

1. Haemocoel: The haemocoel of cockroach is divided into three sinuses by two muscular, horizontal membranes, called **dorsal diaphragm** or 'pericardial septum' and **ventral diaphragm**. Both the diaphragms have valvular pores. There is a series of paired triangular muscles, called **alary muscles**. Every segment has one pair of these muscles situated on the lateral sides of the body. These are attached to the pericardial septum by their broad bases and to the terga by their pointed ends or apices. The three sinuses of the haemocoel are known as **pericardial haemocoel** or the '**dorsal sinus**', the **perivisceral haemocoel** or the '**middle sinus**' and **sternal haemocoel** or '**ventral sinus**' or '**perineural sinus**'. The middle sinus is very large as it encloses most of the visceral organs. The dorsal and ventral sinuses are small in size and they enclose **heart** and **nerve cord**, respectively.

2. Heart

The heart lies in the pericardial haemocoel or dorsal sinus. It is a long, muscular, contractile tube found along the mid dorsal line, beneath the terga of the thorax and abdomen. It consists of 13 chambers. Every chamber opens into the other present in front of it through valvular openings. Three of the thirteen chambers are situated in the thorax and ten in the abdomen. Its posterior end is closed while the anterior end

is continued forward as the anterior **aorta**. At the posterior side of each chamber, except the last, there is a pair of small apertures called '**ostia**' one on each side. Ostia have valves which allow the blood to pass only into the heart from the dorsal sinus.

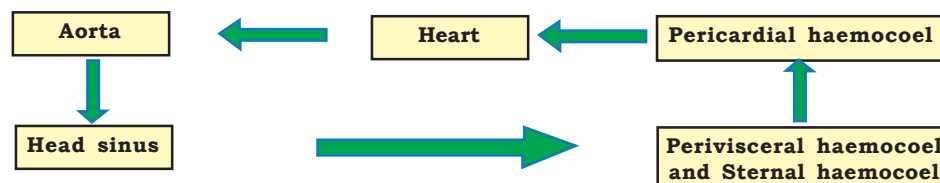
3. Circulatory fluid

The circulatory fluid of *Periplaneta* is colourless and is called **haemolymph**. It consists of a fluid called plasma, and free corpuscles or haemocytes, which are 'phagocytic'. The phagocytes are large in size and can 'ingest' foreign particles such as bacteria. There is no respiratory pigment in the blood and so it plays no major role in respiration. The important functions of the blood are:

1. It absorbs digested food from the alimentary canal and distributes it to the rest of the body.
2. It brings nitrogenous wastes from all parts of the body to the excretory organs for their elimination.
3. It carries defensive phagocytes to the places of infection where they engulf the germs and disintegrating tissue parts.
4. It transports secretions of the ductless glands to the target organs.

4. Circulation of blood

The blood flows forward in the heart by the contractions of its chambers. At the anterior end of the heart, the blood flows into the aorta and from there it enters the sinus of the head. From the head sinus, the blood flows into the perivisceral and sternal sinuses. On contraction of the alary muscles the pericardial septum is pulled down. This increases the volume of the pericardial sinus. Hence blood flows from the perivisceral sinus into the pericardial sinus through the apertures of the pericardial septum. On relaxation of the alary muscles, the pericardial septum moves upwards to its original position. This forces the blood, to enter the chambers of the heart through the ostia from the pericardial sinus.



Course of circulation of blood in *Periplaneta*

7.6 RESPIRATORY SYSTEM OF *PERIPLANETA*

Due to the absence of respiratory pigment, the blood of cockroach is colourless and it cannot carry oxygen to different tissues. Therefore a tracheal system is developed to carry the air directly to the tissues. The respiratory system of cockroach consists of stigmata, tracheae and tracheoles.

1. Stigmata or spiracles: The tracheal system communicates with the exterior by ten pairs of openings called **stigmata** or **spiracles**. The first two pairs of spiracles are present in the thoracic segments, one pair in the mesothorax and one pair in the metathorax. The remaining eight pairs are present in the first eight abdominal segments. Spiracles are located in the pleura of their respective segments. The respiratory system in insects is classified on the basis of number and nature of spiracles. The respiratory system of cockroach is **polypneustic** (as there are more than 3 pairs of spiracles) and the spiracles are **holopneustic** (as all of them are functional). All spiracles are valvular and each of them is surrounded by a chitinous ring called **peritreme**. All spiracles bear small hair like structures called **trichomes** to filter the dust particles. Each spiracle opens into a small chamber called **atrium**.

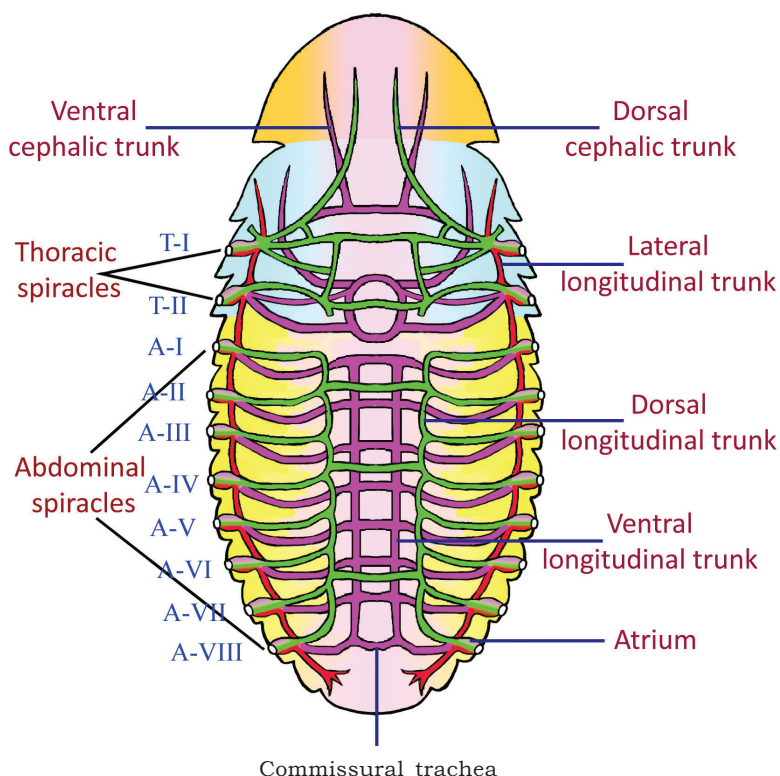


Fig. 7.7 *Periplaneta* : Tracheal system

2. Tracheae: From the atrium of each **thoracic spiracle** several horizontal tracheae run inside. They join with each other in the thorax to form many tracheal trunks like dorsal cephalic, ventral cephalic trunks and their branches. These branches enter all organs of the head. The thoracic region contains lateral longitudinal trunks also. The **abdominal spiracles** lead into atria. From the atrium of each abdominal spiracle three tracheal tubes arise. All these tracheal tubes of one side open into three separate longitudinal tracheal trunks. They are **lateral**, **dorsal** and **ventral longitudinal trunks**. Lateral longitudinal trunks are the longest tracheal trunks. The three pairs of longitudinal tracheal trunks of both the sides are interconnected by many **commissural tracheae**. From all the tracheal trunks several branches are given out, and they enter different organs. Each tracheal branch ends in a special cell called **tracheole cell**.

The wall of the tracheae is made of three layers. They are an outer **basement membrane**, a middle one cell thick **epithelium** and an inner layer of cuticle called **intima**. The intima is produced into **spiral** thickenings called **taenidia**. The taenidia keep the tracheae always open and prevent it from collapsing.

2. Tracheoles

The terminal cell of trachea is called **tracheoblast** or **tracheole cell**. It has several intracellular tubular extensions called **tracheoles**. Tracheoles are devoid of intima and taenidia. They are formed of a protein called **trachein**. Tracheolar fluid is present inside the tracheoles. The level of the tracheolar fluid varies with the metabolic activity of the insect. It is more when the insect is inactive and completely reabsorbed into the tissues, when the insect is more active. Tracheoles reach tissues/ cells and are intimately associated with mitochondria of the cells (to supply oxygen to them).

Which structures prevent the tracheae from collapsing?

4. Mechanism of respiration

Respiration includes two events, viz., inspiration and expiration. The muscles helpful are **dorsoventral muscles** and **ventral longitudinal muscles**. Dorsoventral muscles are the principal muscles of respiration.

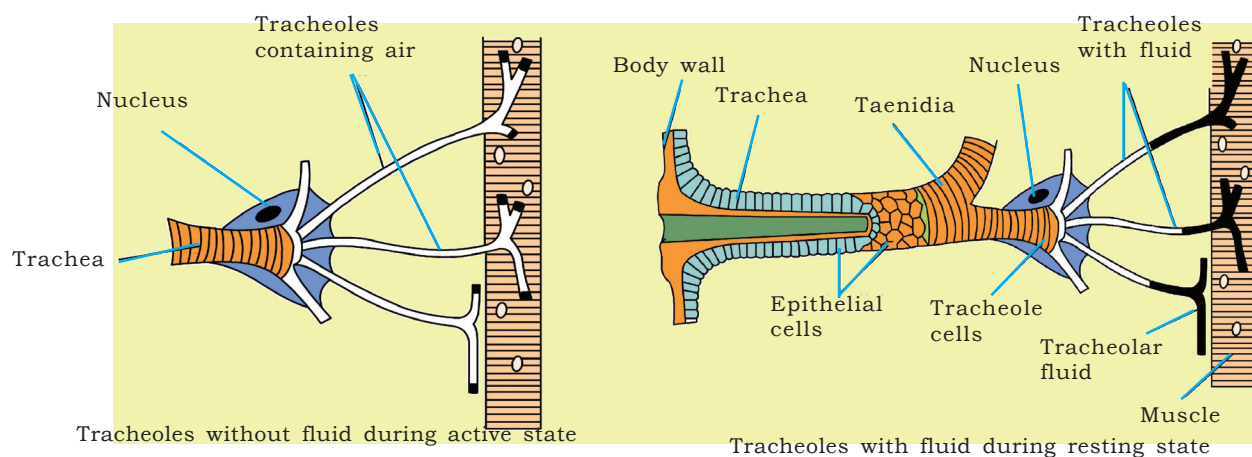


Fig. 7.8 *Periplaneta* : Tracheoles showing movement of tracheolar fluid during respiration

A. Inspiration

Taking in of air is called **inspiration**. It is effected by the relaxation of the dorsoventral muscles and ventral longitudinal muscles. Due to the relaxation of the dorsoventral muscles, tergal plates are elevated and the volume of the body cavity increases. Due to the relaxation of the ventral longitudinal muscles, the telescoped segments come to the **normal position**. So the volume of the body cavity increases in the longitudinal axis. As air is drawn in due to the relaxation of the muscles, the process is a 'passive' process. During inspiration the thoracic spiracles are kept open and the abdominal spiracles are kept closed.

B. Expiration

Sending out air from the body is called **expiration**. On contraction the dorsoventral muscles depress the tergal plates. Body cavity decreases in size and pressure increases. Due to the contraction of the ventral longitudinal muscles, the segments are telescoped and the volume of the body cavity decreases in the longitudinal axis increasing the pressure further. As this process involves the contraction of muscles, expiration is described as an active process. During expiration thoracic spiracles are closed and abdominal spiracles are kept open.

5. Exchange of gases

As air enters the tracheoles, oxygen from the air is taken into the cells and CO₂ is released into haemolymph. The CO₂ from the haemolymph mostly goes out through the inter-segmental membranes of the body wall. Cockroach and some other insects such as grasshoppers and beetles exhibit the phenomenon of **discontinuous ventilation**. In this mode of respiration continuous exchange of gases is interrupted by extended periods during which spiracles remain closed. The expulsion of CO₂ from the body occurs **in bursts**, when the spiracles are open. The exchange of gases depends on the metabolic rate and temperature. When air enters the tracheoles, oxygen diffuses faster into the tissues due to its high partial pressure. At the same time the carbon dioxide of tissues, instead of passing into the tracheal system, goes into the haemolymph. Carbon dioxide is carried more quickly into the haemolymph due to its greater solubility in it. This CO₂ accumulates near the spiracles and diffuses into the atrial chambers near the spiracles and goes out in '**bursts**' through the *abdominal spiracles*. **Opening and closing of spiracles is influenced by CO₂ tension in haemolymph and oxygen tension in the tracheae.**

Which factors regulate the opening and closure of spiracles in *Periplaneta*?

7.7 EXCRETORY SYSTEM OF PERIPLANETA

The excretory system of cockroach helps in eliminating the nitrogenous wastes from the body in the form of **uric acid**. So *Periplaneta* is called a **uricotelic** animal. The structures associated with excretory function are: **Malpighian tubules**, **Fat bodies**, **Uricose glands**, **Nephrocytes** (Ref. NCERT) and **Cuticle**.

1. Malpighian tubules : The malpighian tubules are long, unbranched yellowish tubules, attached at the extreme anterior end of the hindgut, lying freely in the haemolymph, but do not open into it, being blind at the free ends. They are 100-150 in number arranged in 6-8 bundles, each bundle having 15-25 tubules. **Marcello Malpighi**, described these tubules and called them **vasa varicosa**. **Meckel** called them **Malpighian tubules**. Each tubule is lined by a single layer of glandular epithelium with a **brush border** on the inner surface. The 'distal portion' of the tubule is **secretory** and the 'proximal part' is **absorptive** in nature.

The glandular cells of the malpighian tubules absorb water, salts, CO_2 and nitrogenous wastes from the haemolymph and secrete them into the lumen of the tubules. The cells of the proximal part of the tubules reabsorb water and certain inorganic salts. By the contraction of the tubules urine is pushed into the ileum. More water is reabsorbed from it, when it moves in to the rectum and almost solid uric acid is excreted along with faecal matter.

The removal of nitrogenous waste material through the alimentary canal helps in complete reabsorption of water from the wastes and formation of dry uric acid. It is an adaptation for **conservation of water** which is very important in terrestrial organisms.

Is sending out nitrogenous wastes in the form of uric acid advantageous to cockroach? If so, how?

2. Fat bodies: Fat body is a lobed white structure. Urate cells present in these bodies are associated with excretion in a way. These cells absorb and store uric acid throughout the life. This is called '**storage excretion**' as they remain stored in the urate cells of the **corpora adiposa**.

3. Uricose glands: Uric acid is stored in **uricose gland** or **utriculi majores** of the mushroom gland in male cockroach. It is discharged during copulation.

4. Cuticle: Some nitrogenous waste materials are deposited on the cuticle and eliminated during moulting.

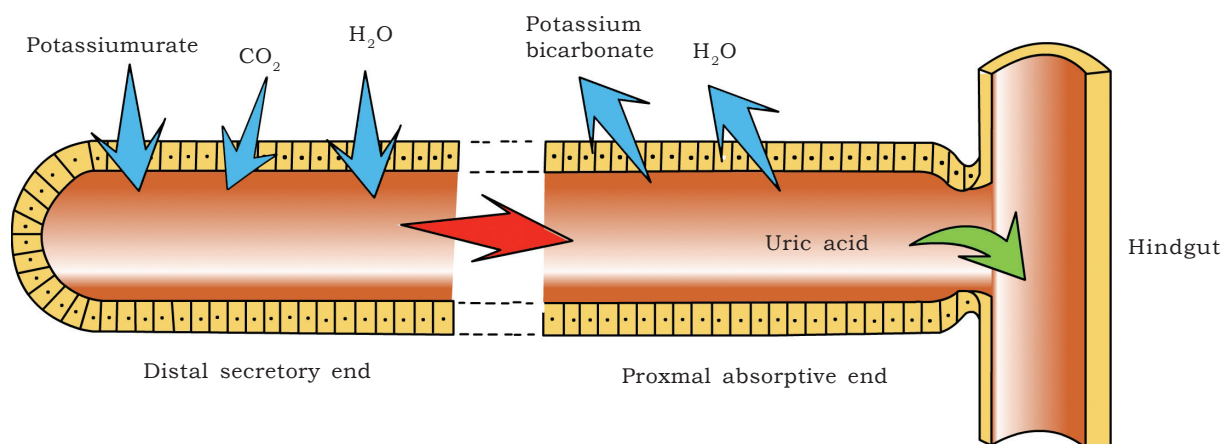


Fig. 7.9 *Periplaneta* : Excretory process in malpighian tubule

7.8 NERVOUS SYSTEM AND SENSE ORGANS OF PERIPLANETA

The nervous system of cockroach consists of central nervous system, peripheral nervous system and autonomous nervous system.

7.8.1 Nervous System

I. Central Nervous System

It consists of a **nerve ring**, and a **ganglionated double ventral nerve cord**.

1. Nerve ring

The nerve ring, which is present around the oesophagus, is formed by the following.

A. Brain

Brain lies above the oesophagus. The brain is mainly a sensory and an endocrine centre. Three lobes of the brain are **protocerebrum**, **deutocerebrum** and **tritocerebrum**. The protocerebrum receives sensory impulses from the compound eyes through optic nerves; deutocerebrum receives sensory impulses from antennae through antennal nerves; and tritocerebrum receives sensory impulses from the labrum. Hence brain is principally '**sensory**' in nature.

B. Sub-oesophageal ganglion

It lies below the oesophagus. It is the principal *motor center*, that controls the movements of **mouthparts**, **legs** and **wings**. It is formed by the fusion of **paired ganglia** of mandibular, maxillary and labial segments of the head.

C. Circum-oesophageal connectives

A pair of circum-oesophageal connectives is present around the oesophagus, connecting the tritocerebrum with the sub-oesophageal ganglion/ sub oesophageal ganglia.

2. Ventral nerve cord

The two ventral nerve cords are solid and ganglionated. They arise from the sub-oesophageal ganglion and extend upto the 7th abdominal segment. The two

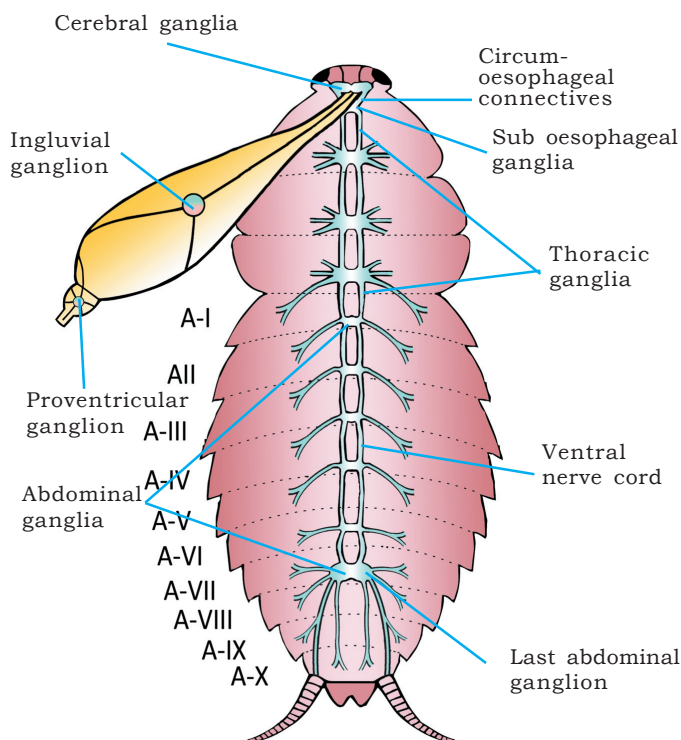


Fig. 7.10 *Periplaneta* : Nervous system

nerve cords remain separate except at the ganglia. Three thoracic ganglia are present, one in each thoracic segment. In addition, there are six abdominal ganglia. The first to the fourth abdominal segments have one abdominal ganglion each. The 5th abdominal segment has no ganglion. The serially 5th abdominal ganglion is present in the 6th segment. The serially 6th abdominal ganglion is present in the 7th segment. The last or the 6th abdominal ganglion is the *largest* of all the abdominal ganglia. It is formed by the fusion of the ganglia of the 7th, 8th, 9th, and 10th abdominal segments.

II. Peripheral Nervous System

It consists of nerves arising from the central nervous system. It receives a pair of, **optic nerves**, from the compound eyes, a pair of **antennal nerves**, from the antennae and a pair of **labral nerves**, from the labrum. Motor neurons of the frontal nerve to the **frontal ganglion** join the sensory neurons of the labral nerve to form the **labro-frontal nerve** arising from the tritocerebrum. Sub-oesophageal ganglion gives off motor nerves to the mandibles, maxillae, labium, wings and legs. It is the principal 'motor centre' in the body.

Thoracic ganglia supply nerves to the parts of their respective segments. Metathoracic ganglia send nerves to the first abdominal segment also. Nerves from the first four abdominal ganglia supply to the organs of the segments 2-6 serially (the 1st to the 4th ganglia innervate segments 2nd to 5th respectively). The 5th ganglion present in the 6th segment innervates the organs of the 6th segment. All organs present in 7th to 10th segments receive nerves from the last abdominal ganglion (present in the 7th segment). The organs include the reproductive organs, copulatory appendages besides anal cerci.

If the sub-oesophageal ganglion is damaged hypothetically, which parts of the cockroach are affected?

III. Autonomous Nervous System

This system is also called **stomatogastric nervous system** or 'visceral nervous system'. It controls the visceral organs, particularly the muscles of the alimentary canal, and the heart. Autonomous nervous system includes four ganglia, a **frontal ganglion** on the dorsal wall of the pharynx, in front of the brain, **hypocerebral ganglion** or occipital ganglion above the oesophagus, behind the brain, a **visceral ganglion** or **ingluvial ganglion** on the wall of the crop and a **proventricular ganglion** on the gizzard. These ganglia contain the 'somata' of the post ganglionic motor neurons. Preganglionic motor neurons of tritocerebrum go to the frontal ganglion as labro-frontal and frontal nerve. Frontal ganglion is connected to the hypocerebral ganglion by a '**recurrent nerve**'. Hypocerebral ganglion is connected to the visceral ganglion and in turn the visceral ganglion is connected to proventricular ganglion.

The movements of which organs in cockroach are controlled by its autonomous nervous system?

7.8.2 Sense organs

In cockroach, various structures like antennae, labrum, maxillary palps, labial palps, compound eyes, simple eyes (**fenestrae**) etc., are sensory in nature. These structures contain various receptors like chemoreceptors, mechanoreceptors, photoreceptors, thermoreceptors, etc. Simple eyes and **ommatidia** of a compound eye are photoreceptors. **Sensillae** are the units of cuticular receptors and chemoreceptors. **Scolopidia** are the subcuticular units of mechanoreceptors of **chordo-tonal organs** (**Johnston's organs** in the antennal pedicels and flagella, **subgenual organs** on the proximal parts of tibiae and **tympanal organs** on the anal cerci; they are sensitive to movement of flagella of antennae, ground vibrations and sound vibrations respectively).

I. Compound eye

A pair of dark kidney shaped compound eyes is present on the dorsolateral sides of the head, one on each side. Each compound eye is composed of about 2000 functional units called **ommatidia**, which are optically independent. The outer surface of the compound eye is divided into about 2000 hexagonal areas called **facets**. Each facet is the outermost portion of a corresponding ommatidium.

1. Structure of an Ommatidium

Each typical ommatidium is an elongated sub unit of the compound eye, consisting of the following parts.

1. **Cornea:** It is the outermost part of an ommatidium and corresponds to a 'hexagonal facet' of the compound eye. It is a biconvex, transparent part of the cuticle and allows light to pass through it. Cornea is secreted by specialized **corneagen cells** of the epidermis. Cornea is a 'refractive' region of an ommatidium.
2. **Corneagen cells or lenticular cells:** These are two transparent specialized epidermal cells that secrete cornea.

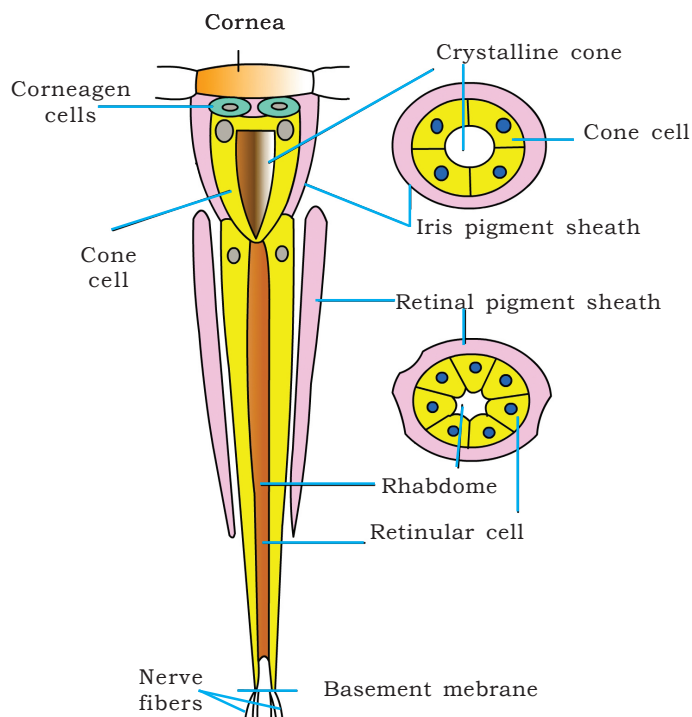


Fig. 7.11 Structure of a typical ommatidium

3. **Vitrellae or cone cells (Semper cells):** These are four transparent more or less conical cells that lie below the corneagen cells. They surround the transparent crystalline cone. Crystalline cone is secreted by the cone cells.
4. **Crystalline cone:** It is the transparent conical structure that is secreted by the vitrellae and is surrounded by them. Light absorbing dark primary pigment cells surround the vitrellae. The region containing the cornea and crystalline cone constitute the **dioptrical** or 'focussing region' of the ommatidium. Crystalline cone focuses the light on to the next part of the ommatidium.
5. **Retinulae:** These are the innermost and elongated cell of an ommatidium. They are seven in number. They rest on the basement membrane. Each cell bears microvilli towards the inner surface. Microvilli of each retinular cell collectively form a **rhabdomere** that contains photoreceptor pigments. These rhabdomeres fuse along the axis of the ommatidium to form the **rhabdome** in the centre. Retinulae are the nerve cells from which sensory nerve fibers leave as the optic nerve to the protocerebrum. They are the **photoreceptor** cells of the ommatidium. Rhabdome and retinulae form the retinal or receptor region. Receptor region is surrounded by seven secondary pigment cells, which absorb light and serve to isolate each ommatidium from the rest (retinal pigment sheath).

2. Functioning of an ommatidium: The position of retinulae below the vitrellae or cone cells, is different in diurnal and nocturnal insects. The two types of images formed in ommatidia of the diurnal and nocturnal insects are the **apposition image** and **superposition image**, respectively.

A. Apposition image: These images are formed in diurnal insects like houseflies in which the retinulae lie immediately below the vitrellae and crystalline cone. These are surrounded by secondary pigment cells, which absorb light rays and prevent them from passing to the adjoining ommatidium. In this type, light rays entering the cornea of a single ommatidium only converge on the rhabdome. Thus, a small part of the visual field is focused on to the receptor part in each ommatidium. The total image formed by the compound eye is a **mosaic** of several **small images**. Such an image is called **apposition image** because it is formed by the juxtaposition of small parts of the visual field. This type of vision is called **mosaic vision**.

B. Superposition image: These images are formed in nocturnal insects like **cockroach** in which the retinulae are present deep below the vitrellae and crystalline cone. Retinal sheath is absent. Therefore, the rhabdome and retinulae of an ommatidium receive not only light rays that enter through its own cornea but also light rays that enter through the corneas of the adjoining ommatidia. This results in the overlapping of images. The image formed by overlapping of images is called '**superposition image**'. It is a '*blurred (nor clear) image*'.

II. Ocelli: At the base of each antenna is a fenestra which represents a simple eye or ocellus. Each ocellus consists of a single corneal facet. These are not involved in image formation but are very sensitive to changes in light intensity.

III. Other sense organs

- Olfactory sensillae** (a type of chemo receptors), which are sensitive to smell, are present on the **maxillary palps**, **labial palps** and **antennae**.
- Gustatory sensillae** (a type of chemoreceptors), which are sensitive to taste, are present on the **labrum**, **maxillary palps** and **labial palps**.
- Thermoreceptor sensillae** (receptors of temperature) occur on the 1st, 2nd, and 3rd segments of the 'tarsi' of the legs.

How is the overlapping of images prevented in diurnal insects?

7.9 REPRODUCTIVE SYSTEM OF PERIPLANETA

Periplaneta is dioecious, or unisexual, and both the sexes have well developed reproductive organs. The sexual dimorphism is evident both externally and internally. The female is different from the male in respect of short and broad abdomen, presence of brood pouch and absence of anal styles.

7.1.2 Male Reproductive System

The male reproductive system consists of a pair of **testes**. These are elongated and lobed structures lying one on each lateral side in the **fourth to sixth abdominal segments**. They are embedded in the fat bodies. From the posterior end of each testis, there starts a thin duct, the **vas deferens**. The two vasa deferentia run backwards and inwards to open into a wide median duct, the **ductus ejaculatorius** in the seventh segment. A characteristic **mushroom shaped gland** is present in the 6th and 7th abdominal segments which functions as an accessory reproductive gland.

The gland consists of two types of tubules: **i)** long slender tubules, the **utriculi majores** or 'peripheral tubules'. **ii)** Short tubules, the **utriculi breviores**. Secretion of utriculi majores forms the inner layer of the spermatophore while that of utriculi breviores nourishes the sperms. These tubules open into the anterior part of the **ejaculatory duct**.

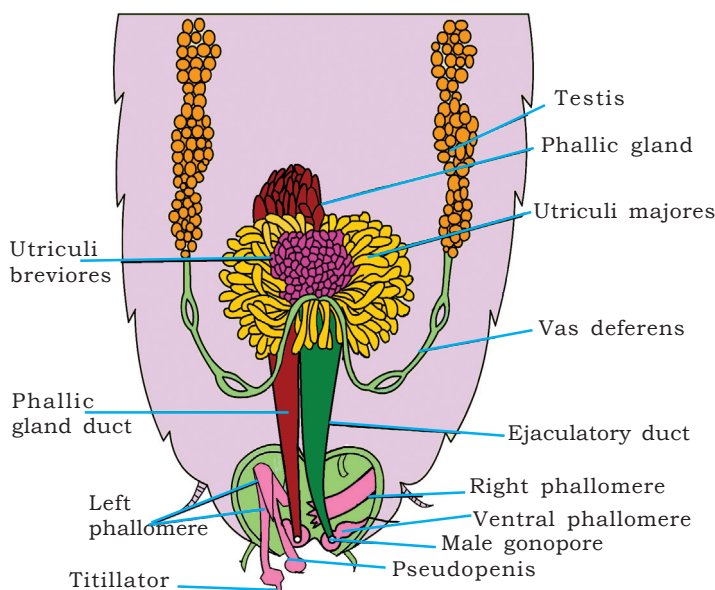


Fig. 7.9 *Periplaneta* : Male reproductive system

The **seminal vesicles** are present on the ventral surface of the ejaculatory duct. These sacs store the sperms in the form of bundles called **spermatophores**. The ejaculatory duct is a muscular tube that extends posteriorly and opens at the **gonopore** or the 'male genital pore' on the ventral phallomere. The duct of **phallic** or **conglobate gland** also opens near the gonopore. Its function is still not known. Surrounding the male genital opening there are three chitinous and asymmetrical structures called **phallic** organs or **gonapophyses** or **phallomeres** which help in copulation. These are the male external genitalia. The term phallomere refers to 'penis'. The left phallomere bears a pseudopenis and titillator.

Note: Ejaculatory duct is believed to secrete the middle layer and the conglobate gland secretes the outer layer of a spermatophore.

7.9.2 Female Reproductive System

The female reproductive system of *Periplaneta* consists of a pair of ovaries, a pair of oviducts, vagina, spermathecae, spermathecal papilla, and colleterial glands.

Ovaries

A pair of large **ovaries** lies laterally in 2 to 6 abdominal segments. They are light yellow in colour surrounded by fat bodies. Each ovary consists of eight tubules called ovarian tubules or **ovarioles**. Each ovariole consists of a tapering anterior filament called **germarium**, and a posterior wider **vitellarium**. The germarium contains various stages of developing ova, and the vitellarium contains mature ova with yolk. The tapering ends of the ovarioles of each ovary unite to form a single thread which attaches to the dorsal body wall. The ovarioles, at their posterior end unite to form a short wide **oviduct**. The oviducts unite to form a very short median **vagina**. The vertical opening of the vagina is called female genital pore. It opens into a large genital pouch on the eighth sternum. A spermatheca or '**receptaculum seminis**', consisting

of a left-sac like and a right filamentous caecum, is present in the 6th segment which opens by a median aperture on a small spermathecal papilla in the dorsal wall of the genital pouch on the ninth sternum. In a fertile female, the spermatheca contains spermatophores, received during copulation.

A pair of branched **colleterial glands** is present behind the ovaries. These glands open into the genital pouch separately, just above the

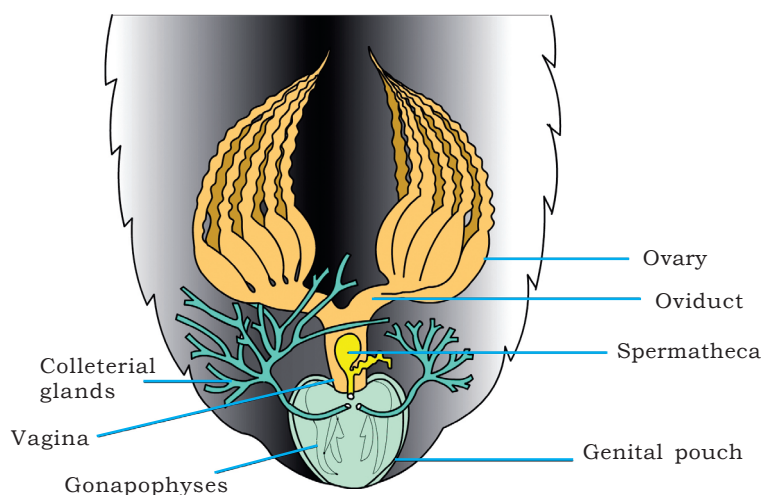


Fig. 7.9 *Periplaneta* : Female reproductive system

spermathecal aperture. Secretion of the two collateral glands forms a hard egg case called **ootheca** around the eggs.

Genital or brood pouch is formed by 7th, 8th, and 9th abdominal sterna. The sternum of the seventh segment is boat shaped and forms the floor and side walls of the genital pouch. The sterna of the eighth and ninth segments, which are tucked into the seventh segment, constitute the anterior wall and the roof of the genital pouch, respectively. The genital pouch has two chambers the anterior 'gynatrium' or **genital chamber** and posterior 'vestibulum' or **oothecal chamber**.

Three pairs of plate like chitinous structures called gonapophyses are present around the female genital aperture. These gonapophyses guide the ova into oothecal chamber acting as ovipositors. These are the *female external genitalia*, which help in copulation also.

7.9.3 Life history

The fertile female produces certain **pheromones** (chemical substances useful in chemocommunication), that attract the males for copulation. Copulation occurs during summer and it takes place during the night time. For copulation the male and female cockroaches bring their posterior ends very close. The male deposits a spermatophore on the spermathecal papilla of the female with the help of its phallomeres.

In *Periplaneta* sixteen eggs mature at a time and are passed into genital chamber. Fertilisation occurs in the gynatrium. The secretions of collateral glands form the egg case or **ootheca** around the fertilized eggs in the oothecal chamber. The ootheca is dark reddish to blackish brown capsule. It contains 16 eggs arranged in two rows of eight each. The oothecae are deposited at a suitable warm and dark place. On an average, female produces 9-10 oothecae. The development of eggs occurs inside the ootheca. Cockroach is **paurometabolous**, which means the development (metamorphosis) is gradual through nymphal stages. The eggs in ootheca undergo cleavage resulting in blastula followed by gastrula. Further development results in the formation of a **nymph**. It (nymph) is a young cockroach hatching from the ootheca. It resembles the adult, except in size, colouration, sexual immaturity and absence of wings. After hatching, the nymphs feed and undergo a series of 13 moultings or ecdyses. Shedding of exoskeleton is called '**moulting**' or '**ecdysis**'. After the last ecdysis the nymph grows into fully formed adult, with wings (the nymphs bear small wing pads).

Note: Moulting permits growth in the size of the nymph.

What do you name the young one of a cockroach which resembles the adult except in size, wings, colour and genitalia?

GLOSSARY

Abdomen: Third or posterior division (tagma) of an insect

Antenna: A sensitive feeler from the animal's head; tactile and olfactory in function

Appendage: A movable projecting part of the body; mouth parts are modified appendages.

Arolium: A soft hairy pad between claws in the leg of cockroach.

Arthropod: A phylum of segmented invertebrates with jointed legs.

Biramous: Possessing two branches e.g. the exopodite and endopodite.

Blastula: Embryonic stage with blastocoel, which is also called the 'primary body cavity'.

Cephalic: Relating or to the head.

Cervix: Neck.

Chitin: A polysaccharide-protein complex; a horny substance.

Copulation: Sexual process/union for the transfer of sperm from one to another of mating partner.

Cursorial: Swift runner.

Cuticle: External skeletal structure of the insect's body; it has epicuticle, mesocuticle and endocuticle.

Dioecious: Species population in which male and female organs occur in separate individuals.

Dioptric region: The part that focuses light rays on to the inner part of an ommatidium .

Ecdysis: Shedding of the outer body layer (cuticle).

Germarium: Tapering anterior filament of an ovariole.

Haemocoel: The body cavity of an arthropod or a mollusk filled with haemolymph; derived from the blastocoels of the embryo ; also called the 'primary body cavity'

Nervures: Tubular hollow network of the wing of a cockroach.

Ootheca: A hard purse like structure containing eggs in two rows.

Ovariole: A single tubule of an ovary containing developing eggs.

Paurometabolic: Gradual development/ metamorphosis by the nymphal stages of certain insects.

Podomere: A segment of an arthropod leg.

Pulvillus: A soft hairy pad between claws.

Sternum (plural: sterna): A ventral sclerite of a body segment of insects such as a cockroach.

Tagma (plural: tagmata): One of the regions of the body of an insect; division of the body into different regions is called tagmatisation.

Tergum (plural: terga) : Dorsal sclerite of the body segment of an insect.

VERY SHORT ANSWER TYPE QUESTIONS

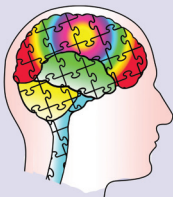
1. What are the structures with which cockroach walks on smooth surfaces and on rough surfaces respectively?
2. How is a tripod formed with reference to locomotion in cockroach?
3. Name the muscles that help in elevating and depressing the wings of a cockroach?
4. Name the different blood sinuses in cockroach?
5. How are the fat bodies similar to the liver of the vertebrates?
6. Which part of the gut secretes the peritrophic membrane in cockroach?
7. In which part of the gut of cockroach, water is reabsorbed?
8. Write the names of mouthparts in cockroach that help in biting and tasting the food.
9. What are alary muscles?
10. What is haemocoel?
11. Why is the blood of *Periplaneta* called haemolymph?
12. What is the function of haemocytes found in the blood of *Periplaneta*?
13. Write important functions of blood in *Periplaneta*?
14. What are trichomes? Write their functions.
15. Why is the respiratory system of cockroach called polypneustic and holopneustic system?
16. What is intima?
17. During inspiration which spiracles are kept open and which are kept closed?
18. Which factors regulate the opening of the spiracles?
19. The nitrogenous wastes in *Periplaneta* are removed from the body through alimentary canal. Why?
20. How do fat bodies help in excretion?
21. Which structure of cockroach acts as sensory and endocrine centre?
22. Distinguish between scolopidia and sensillae.
23. Which of the abdominal ganglia is the largest and why?
24. Why is the brain called the principal sensory centre in cockroach?
25. Distinguish between apposition image and superposition image.
26. List out the characters that help in understanding the difference between male and female cockroaches.
27. What is the function of mushroom gland in cockroach?
28. Compare the utriculi majores and utriculi breviores of the mushroom gland functionally.
29. What are gonapophyses?
30. How is colleterial gland helpful in reproduction of *Periplaneta*?

**SHORT ANSWER TYPE
QUESTIONS**

1. Draw a neat labeled diagram of the mouthparts of cockroach?
2. Describe the physiology of digestion in cockroach?
3. Draw a neat labeled diagram of the salivary apparatus of cockroach?
4. Describe the structure and function of the heart in *Periplaneta*?
5. Describe the process of blood circulation in *Periplaneta*?
6. What are different excretory organs in *Periplaneta*? Describe the process of excretion in detail.
7. How does *Periplaneta* conserve water? Explain it with the help of excretion in it?
8. Draw a neat and labeled diagram of ommatidium.
9. Describe the male reproductive system of cockroach.
10. Describe the female reproductive system of cockroach?

LONG ANSWER TYPE QUESTIONS

1. Describe the digestive system of cockroach with the help of a neat labelled diagram?
2. Describe the blood circulatory system of *Periplaneta* in detail and draw a neat and labelled diagram of it?
3. Describe the respiratory system of cockroach with the help of neat and labelled diagrams?
4. Describe the reproductive system of *Periplaneta* and draw neat and labeled diagrams of it.



For ignited minds

- NOT FOR EVALUATION

1. Do you know that animals send out nitrogenous wastes in the form of ammonia, urea, uric acid, guanine, etc. Ammonia is the **most toxic substance** and it can be sent only in very dilute form. It requires loss of more water (urine is hypotonic). Some fishes excrete ammonia (ammonotelic), as availability of water is not a problem for them. Urea is relatively less toxic and it can be sent out with relatively less water. Frog, earthworm etc., send out wastes mostly in the form of urea (ureotelic).

Note: Do you know that urea is formed mostly in our liver via the ornithine cycle? Uric acid is the least toxic substance and so it can be sent out in a concentrated form. So, animals such as reptiles, birds and insects for which conservation of water is very important send the wastes in the form of semisolid uric acid crystals (uric acid is insoluble in water, unlike ammonia and urea). Aquatic reptiles such as turtles excrete 'urea' like the mammals. However urine cannot be concentrated due to lack of Henle's loops in their nephrons. Now list out various types of animals depending on the nitrogenous waste excreted and state reasons for their nature.

2. When prawn, scorpion etc., have an oxygen carrying pigment in their haemolymph, why is it not seen in the blood of cockroach? Think of it.
3. When the labrum is removed or labral nerves are cut off, cockroach does not lose its ability/sense of tasting food. Why?

CLUE – Recall: functions of all appendages.

4. Why do insects moult in the early part of their life?

CLUE: It has something to do with growth.

5. The compound eye of cockroach, in a way, is a better adaptation for survival, than the human eye. Do you agree with the statement? If so how do you substantiate? **CLUE:** As a human being,

you can concentrate on only a particular part of your visual field only at a time, even though your visual field is much more larger/ wider.

6. What happens if the serially fifth abdominal ganglion is hypothetically damaged in a cockroach? The functioning of organs of which abdominal segment is affected?
7. What do you call the units of the mechanoreceptors in organ such as anal cerci?
8. If 20% of the ommatidia are damaged in each compound eye, a diurnal insect can see only 80% of the visual field. Do you agree with the statement? If so explain.
9. Why is storage of uric acid in corpora adiposa called storage excretion?
10. Of the following organs – antennae, mandibles, maxillae, legs and wings, which do you think is taxonomically significant when we think of insect such as a cockroach?
11. If the labro-frontal nerve is severed (cut off), will it affect the circulation of blood? How?
12. If the seminal receptacle of a cockroach is removed, and if copulation occurs soon after its removal what will be the effect on the reproductive ability of that cockroach?
13. The heart of a chordate is said to be myogenic (which means contraction is initiated in itself). If the heart is removed from the body it will keep beating provided some nourishing liquids such as Ringers solution is pumped through it. Will the same thing happen with the heart of the cockroach? Explain.
14. You know striated muscles appeared for the first time in the arthropods during evolution. Muscles need endo-skeletal parts for attachment. Cockroach does not have an endo-skeleton comparable to that of a chordate. Do you know how this problem of attachment surfaces is solved in cockroach? (Clue: Cuticle of cockroach shows inward extensions such as apodemes).
15. If the mesothoracic wings such as apodemes of a cockroach are removed, do you think it can still fly? If so, give reasons.